EPIC-Rec: Evidence-Preserving Counterfactual Explanations for Top-K Graph Recommendation

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Abstract

Counterfactual explanations for recommender systems name a small set of past interactions whose removal changes a recommendation. Existing methods remove the target item but leave the rest of the list to chance, and they do not ask whether the reasons behind the remaining recommendations survive the edit. We study the stricter question: can one recommendation be removed with a minimal edit to the user's history while every other top-K item stays in the list and keeps at least one of its original, score-contributing evidence paths? We formulate this as a constrained minimum-mask problem with three conditions—removal (C1), membership preservation (C2) and evidence preservation (C3)—and introduce EPIC-Rec (Evidence-Preserving Interaction Counterfactuals for Recommendation). On an inductive, additive path-scoring recommender the problem becomes a mixed-integer linear program that we solve exactly, with a small score margin, using lazy constraint generation, and every optimality or infeasibility claim is re-checked under a second solver configuration. On MovieLens-1M, across 2,000 user–target pairs from 200 users, a mask that removes the target exists for 100.00% of pairs, a mask that also keeps the other $K-1$ items for 83.30% [80.92, 85.75], and one that also keeps their evidence for 73.10% [70.11, 76.05]. On Last.fm the same three rates are 95.70% [94.75, 96.60], 30.05% [28.05, 31.90] and 28.05% [26.15, 30.00]. A target-score order that stops at removal keeps the list and the evidence in only 27.5% of the MovieLens cases, and 12.4% of the Last.fm cases, in which it removes the target. Surgical counterfactuals are common on MovieLens and uncommon on Last.fm, and the exact formulation makes that entanglement measurable.

1 Introduction

Counterfactual explanations justify an individual algorithmic decision by naming a small change to the input that would have produced a different output [27]. In recommendation, the input a user controls is their interaction history, and an explanation takes the form "had you not watched X and Y, you would not have been recommended Z" [10, 26]. Recent work has extended such explanations to graph-based recommenders [6, 7, 18], to learned and attribution-based explainers [3, 4, 11], to sequential models [16], and to evaluation protocols that look beyond the top-1 item [19, 21]. The closest work, CoSRec [32], learns sparse interaction masks for a fixed graph recommender, verifies them by recomputing the ranking, and reports how much of the surrounding list remains stable.

What these methods do not provide is a *guarantee* about the rest of the list. A mask that removes the target may also push other items out, and it may destroy the evidence that justified them. Stability is measured after the fact, not enforced. For a user who wants one unwanted

recommendation gone, or for an auditor who wants to know whether a recommendation can be explained independently of its neighbours, the relevant question is stricter:

> *Can we remove a specific recommendation with a minimal edit to the user's history while preserving every other top-K item and at least one of its original, score-contributing evidence paths?*

Why would one recommendation need to be removed without disturbing the others? Three situations motivate the question. First, user control: a user who presses "not interested" on one item—a title watched for a child, a one-off gift purchase, a product already bought elsewhere—expects that item to disappear, not the rest of a list they were satisfied with. Second, privacy: a single recommendation can reveal a sensitive inference about the user, and removing it should not degrade unrelated suggestions. Third, explanation and audit: even when nothing is actually removed, a minimal counterfactual ("had you not interacted with A and C, T would not have been recommended") is a verifiable account of why an item was recommended, and transparency obligations for recommender systems, such as those in the EU Digital Services Act [8], ask platforms to explain the main parameters of their recommendations and to offer users control over them. When surgical removal is impossible, that fact is itself informative: it shows that the recommendation is entangled with others in the list and cannot be explained or withdrawn in isolation.

We make four contributions.

1. A problem formulation with a joint guarantee: target removal (C1), exact top-K membership preservation for the other $K - 1$ items (C2), and survival of at least one original, score-contributing evidence path for each of them (C3), with an explicit ranking rule and an explicit evidence unit (Section 3).

2. An exact solution method. On an inductive, additive path scorer every score is linear in the binary mask, and a margin version of the problem becomes a mixed-integer linear program (MILP) that we solve with lazy constraint generation; C3 reduces to covering constraints (Section 4). We measure how often the margin version and the original problem differ.

3. A measured cost of each constraint on MovieLens-1M and Last.fm: how many user–target pairs admit a mask, how many are proven infeasible, and the paired increase in minimum mask size from C1 to C1 + C2 to C1 + C2 + C3, stratified by target rank and history length (Section 6).

4. Evidence that constraint-unaware heuristics rarely satisfy C2 and C3 by chance, and a comparison of heuristics and beam search with the exact optimum.

The counterfactual edge-mask mechanism builds on our earlier work on counterfactual inverse link prediction (CILP) [15], which removes edges from unipartite social graphs. EPIC-Rec moves this idea to user–item–entity graphs, replaces a single link prediction with a ranked list, and adds the evidence constraint. Figure 1 illustrates the problem and the method.

2 Related work

History-based counterfactual explainers. PRINCE [10] finds minimal sets of user actions that replace the top recommendation of a personalised-PageRank recommender over a heterogeneous graph. We do not run PRINCE at the scale of this study: it explains that recommender rather than the path-additive backbone, and no comparison under C2 or C3 is claimed. ACCENT [26] extends

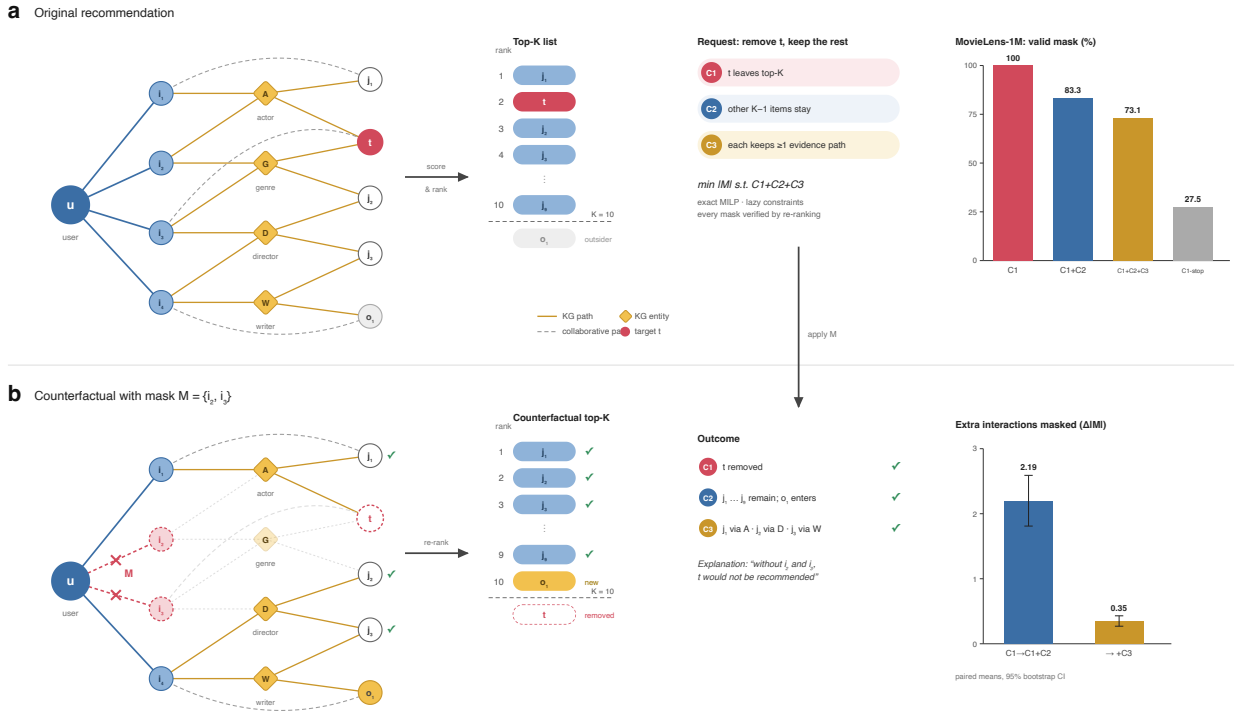


Figure 1: **EPIC-Rec removes one recommendation while keeping the rest of the top-K list and its evidence.** **a**, User u's history (i_1–i_4) reaches candidate items through knowledge-graph entities (diamonds; solid lines) and collaborative paths (dashed). Scores are sums of path contributions, giving a top-K list ($K = 10$) that contains the target t; o_1 is the best item outside the list. EPIC-Rec looks for the smallest set of history interactions M to mask such that t leaves the list (C1), the other $K - 1$ items stay (C2), and each keeps at least one of its original evidence paths (C3). The problem is solved exactly as a mixed-integer linear program and every mask is checked by re-ranking. Right: MovieLens-1M, share of 2,000 pairs for which a valid mask exists (100.00%, 83.30% and 73.10% for C1, C1 + C2 and C1 + C2 + C3). The grey bar is the share of target-score edits that stop at removal and still keep the list and the evidence (27.5%). Last.fm is in Table 2. **b**, Masking $M = \{i_2, i_3\}$ removes their paths (faded). t leaves the list, j_1–j_9 remain, each supported by a surviving path (✓), and o_1 takes the free slot. Right: paired increase in minimum mask size on MovieLens when C2 is added to C1 (1.348 [1.185, 1.524]) and when C3 is then added (0.313 [0.250, 0.377]); means with 95% user-level bootstrap CI; $\alpha = 0.5$.

influence functions to pairs of items and removes the user's most influential interactions until the top recommendation is replaced. LXR [3, 11] trains an explainer network with counterfactual loss terms that scores the relevance of each piece of user data to a recommended item; the journal version adds per-explanation fine-tuning at inference time. SPINRec [4] computes path-integrated attributions for implicit-feedback data, sampling several plausible user profiles as baselines and keeping the most faithful integration path. These methods explain one recommended item; they do not constrain the rest of the list.

Graph explainers for recommendation. GREASE [6] trains a surrogate GNN around the user–item pair and optimises perturbations of its adjacency matrix to obtain factual and counterfactual explanations; it has been extended to top-k recommendation [7]. GNNUERS [18] perturbs the user–item graph to explain unfairness, and CF-GNNExplainer [17] removes edges to flip node predictions. CoSRec [32] learns a sparse retention mask over a user's interactions for a frozen LightGCN [13] with a ranking loss and a sparsity penalty, verifies the change by recomputation, and measures list stability. EPIC-Rec differs by enforcing membership as a hard constraint and by requiring evidence to survive.

Path-based explanations. Semantic knowledge has long been used to enrich usage-based recommendation [1], and knowledge-graph recommenders such as KGAT [30] and RippleNet [28] propagate preferences along entity paths. CERec [31] uses reinforced path reasoning to generate attribute-level counterfactuals over a knowledge graph, PEARLM [2] constrains path generation to paths that exist in the graph, and CountER [25] perturbs item aspects. None of them asks whether the paths that justify *other* recommendations survive a counterfactual edit.

Sequential and list-level evaluation. TRACE [16] produces training-free, ranking-aware bring-in and push-out counterfactuals for sequential recommenders from score queries. CFExpRec [21] and Beyond Top-1 [19, 20] show that evaluating explanations at top-1 gives inconsistent conclusions and propose top-K protocols. We follow this list-level view, but as a constraint on the explanation rather than only as an evaluation metric.

3 Problem formulation

Setting and ranking rule. User u has training history H_u. A fixed recommender scores every item of the candidate pool C_u, the catalogue minus H_u. For a mask $M \subseteq H_u$ let $s_M(j)$ be the score of j computed from $H_u \setminus M$. Items of C_u are ranked by s_M in descending order, *ties broken by item identifier in ascending order*; top-$K(M)$ is the first K items of this order. This one rule is used by the recommender, by the verifier and by every search method. $L_u = \text{top-}K(\emptyset)$ is the original list, $t \in L_u$ the target, $R = L_u \setminus \{t\}$ the retained items and $O = C_u \setminus L_u$ the outsiders.

Mask domain. The mask is restricted to $D_u(t) = \{i \in H_u : A_{it} > 0\}$, the interactions with at least one path to t (Section 4), so that an explanation names only interactions that support the target. On MovieLens-1M this restriction removes nothing: $D_u(t)$ covers 100.0% of the history, because collaborative paths connect every scored pair. On Last.fm it covers 70.3%.

Evidence unit: the individual path. The unit of evidence is one path $p = (i, \rho, x, j)$ from an interaction $i \in H_u$ through relation ρ and intermediate node x to item j. For a knowledge-graph relation, x is an entity that i and j share through ρ ($i \xrightarrow{\rho} x \xleftarrow{\rho} j$); for the collaborative relation, x is

another user who interacted with both i and j in the training data ($i \leftarrow x \rightarrow j$). The path's origin is $e(p) = i$; the path survives M when $e(p) \notin M$. Its contribution $c(p)$ is the amount it adds to $s(j)$ (Section 4); on our backbone it does not depend on what else is masked. Two paths that share an origin and a relation but pass through different entities or users are different pieces of evidence.

Definition 1 (Original evidence set)**.** *For a retained item $j \in R$ let $\tau_j = \alpha \max_{p \to j} c(p)$, with $\alpha \in (0, 1]$ and the maximum taken over all paths from H_u to j. The original evidence set of j is $P_j = \{p \to j : c(p) \geq \tau_j\}$.*

The threshold is relative to the item's strongest single path, so the same α is comparable across users and items. A retained item with no path from H_u has an empty evidence set and cannot satisfy C3; such items make setting C infeasible for the pair, and we report how often this happens.

Definition 2 (Problem P)**.** *Find $M \subseteq D_u(t)$ of minimum size such that*

C1 (removal)	$t \notin \text{top-}K(M)$,
C2 (membership)	$R \subseteq \text{top-}K(M)$,
C3 (evidence)	$\forall j \in R\ \exists p \in P_j :\ e(p) \notin M.$

Setting A asks for C1 only, setting B for C1+C2, and setting C for all three.

Retained items may reorder, and under C1+C2 exactly one outsider takes the freed position.

Margin variant P_ε. A MILP cannot express "ranked above, ties broken by identifier" without strict inequalities, so the solver works on a margin variant in which every ordering the constraints need holds with a gap of at least $\varepsilon = 10^{-6}$ in masked score:

$\mathbf{A}_\varepsilon$	$\big\lvert\{c \in C_u \setminus \{t\} : s_M(c) \geq s_M(t) + \varepsilon\}\big\rvert \geq K,$
$\mathbf{B}_\varepsilon$	$\exists r \in O :\ s_M(r) \geq s_M(t) + \varepsilon$ and, for every $j \in R$, $s_M(j) \geq s_M(t) + \varepsilon$ and $s_M(j) \geq s_M(c) + \varepsilon\ \ \forall c \in O \setminus \{r\},$
$\mathbf{C}_\varepsilon$	$\mathbf{B}_\varepsilon$ and C3.

Proposition 1. *Every mask feasible for P_ε is feasible for P in the same setting. Hence the P_ε optimum is an upper bound on the P optimum. The two problems can differ only when a mask needs an ordering with a gap in $[0, \varepsilon)$, in particular an exact tie resolved by item identifier.*

Proof sketch. Under A_ε, K items score strictly above t, so t is not in the first K whatever the tie rule. Under B_ε, each retained item is beaten at most by other retained items and by r, so it ranks within the first K; and t is beaten by the $K - 1$ retained items and by r. C3 does not involve the ranking. □

A proof of infeasibility for P_ε is therefore not a proof for P, and a P_ε optimum can exceed the P optimum. We measure both effects by exhaustive search on every pair with $|D_u(t)| \leq 14$ in a separate MovieLens sample (Section 6.2). The main samples are not enumerated.

Scope of strict preservation. Related items—a sequel, a film by the same director—often share the target's evidence, and a user may expect them to disappear too. Strict $K - 1$ preservation is therefore used in two roles: as a diagnostic that measures how entangled a recommendation is with the rest of the list, and as a single-item filter for users who endorse the remaining items.

4 Method

4.1 An inductive, additive path backbone

Study 1 uses a recommender whose score is an explicit sum over evidence paths:

$$s(u,j) = \beta \log(1 + \mathrm{pop}_j) + \frac{1}{|H_u|} \sum_{i \in H_u} \sum_{\rho \in \mathcal{R}} \theta_\rho S_\rho[i,j], \tag{1}$$

where for a knowledge-graph relation ρ $S_\rho[i,j] = \sum_x 1/\deg_\rho(x)$ over the entities x that i and j share through ρ, and for the collaborative relation $S_{\mathrm{co}}[i,j] = \mathrm{co}(i,j)/\sqrt{\mathrm{pop}_i\,\mathrm{pop}_j}$, one term per co-interacting user. Each summand is one path p with contribution $c(p) = \theta_\rho/(\deg_\rho(x)|H_u|)$ or $c(p) = \theta_{\mathrm{co}}/(\sqrt{\mathrm{pop}_i \mathrm{pop}_j}\,|H_u|)$; the normalisations are frozen at their pre-mask values. Co-occurrence counts and popularity are computed on the training split. MovieLens keeps the 11 Satori film relations that survive the frequency filter; Last.fm keeps the 8 music relations released with KGCN [29]. Collaborative and trust-based signals remain central to recommendation [9]; our backbone represents the collaborative signal as explicit co-interaction paths, so it enters the evidence sets like any knowledge-graph relation. There is no user embedding, so masking removes all of M's signal and no retraining is needed. The weights $\theta_\rho \geq 0$ and $\beta \geq 0$ are learned with BPR [22] under non-negativity constraints on leave-one-out features: for a training triple (u,p,n) the model input is $H_u \setminus \{p\}$, and the co-occurrence and popularity statistics that contain u's own interaction with p are corrected accordingly.

Two consequences make the problem tractable. Masking interaction i subtracts a fixed row, $s_M(j) = s(j) - \sum_{i \in M} A_{ij}$ with $A_{ij} = W_{ij}/|H_u|$ and $W = \sum_\rho \theta_\rho S_\rho$, so every score is linear in the mask. And a surviving path's contribution does not depend on the mask, so with $P_{ij} = \max_{p:\, e(p)=i,\, p \to j} c(p)$ the largest single-path contribution from i to j, C3 holds for j if and only if some interaction in $Q_j = \{i : P_{ij} \geq \tau_j\}$ is unmasked.

4.2 Exact MILP for P_ε

Let $m_i \in \{0,1\}$ indicate $i \in M$ for $i \in D_u(t)$ and $z_r \in \{0,1\}$ select the replacement among a candidate set $\mathcal{O} \subseteq O$. Writing $\Delta_{ab}(m) = s(a) - s(b) - \sum_i m_i(A_{ia} - A_{ib})$, setting C is

$$\begin{aligned}
\min\ & \textstyle\sum_i m_i \\
\text{s.t.}\ & \textstyle\sum_{r \in \mathcal{O}} z_r = 1, \qquad \Delta_{rt}(m) \geq \varepsilon - B_{rt}(1 - z_r)\ \ \forall r \in \mathcal{O}, && (2)\\
& \Delta_{jt}(m) \geq \varepsilon\ \ \forall j \in R, && (3)\\
& \Delta_{jc}(m) \geq \varepsilon - B_{jc} z_c\ \ \forall j \in R,\ c \in O\ \ (z_c \equiv 0 \text{ for } c \notin \mathcal{O}), && (4)\\
& \textstyle\sum_{i \in Q_j}(1 - m_i) \geq 1\ \ \forall j \in R. && (5)
\end{aligned}$$

Setting B drops (5); setting A replaces the ordering constraints by indicators y_c, one per candidate, and requires $\sum_c y_c \geq K$. Each big-M constant is the tightest valid value for its row, $B_{ab} = \max(0, \varepsilon - \min_m \Delta_{ab}(m))$.

Replacement pruning. Let $\mathrm{lb}(c) = s(c) - \sum_{i \in D_u(t)} A_{ic}$ be the lowest score c can reach. The replacement must satisfy $s(r) \geq \mathrm{lb}(t) + \varepsilon$, and we may further restrict it to $s(r) \geq \max_{c \in O} \mathrm{lb}(c)$. The reason: if a mask is feasible with replacement r', it is also feasible with b, the best outsider under the masked scores, because $s_M(b) \geq s_M(r') \geq s_M(t) + \varepsilon$ and every retained item already beats b (and hence r') by ε; and $s(b) \geq s_M(b) \geq s_M(c) \geq \mathrm{lb}(c)$ for every outsider c. The rule never excludes all candidates of a feasible instance; the regression tests compare the solver with pruning switched on and off.

Lazy constraints and outcomes. The threat constraints (4) are dense and numerous. We start with the 40 highest-scoring outsiders, solve, recompute the ranking under the returned mask, add every outsider that violates B_ε, and repeat. Every intermediate model is a relaxation, so an infeasible relaxation proves infeasibility and a verified optimum of a relaxation is optimal. Each instance has a budget of 20 seconds of wall-clock time, including model building; the solver is HiGHS [14] with a zero relative MIP gap and feasibility tolerances of 10^{-9}. Every instance ends in one of four outcomes: *optimal* (verified mask and matching lower bound), *feasible* (verified mask, optimality not proven within the budget), *infeasible* (proven for P_ε), or *timeout* (neither). A mask *exists* for the pair when the outcome is optimal or feasible.

Certification. A claim of optimality or infeasibility rests on one branch-and-bound run. On one MovieLens pair the study configuration (presolve off for C1 + C2 and C1 + C2 + C3, feasibility tolerances 10^{-9}) returned a non-minimal mask as optimal. Every such claim is therefore re-solved for 60 seconds under a second HiGHS configuration, presolve on with primal tolerance 10^{-7} and MIP tolerance 10^{-6}, on the model that would refute it: an optimum of size k with $|M| \leq k-1$, an infeasibility claim with no size cap. Infeasible confirms the claim. A returned mask replaces the result only after the independent verifier accepts it, and the new size is certified again. Anything else is undecided: a claimed optimum is reported as feasible, and a claimed infeasibility as a timeout. Optima of size 0 and infeasibility found by the structural pre-checks involve no search. The problem definitions are unchanged. On the MovieLens main sample this confirmed 7790 of the 8,000 exact results, replaced one optimum of size 4 by a verified optimum of size 3, and left 28 claimed optima unconfirmed. On Last.fm all 8000 exact results were confirmed.

4.3 Exhaustive search, beam search and heuristics

Exhaustive search. The main samples are not enumerated. On the margin sample, when $|D_u(t)| \leq 14$, we enumerate masks in order of size and return the first feasible one, once under P and once under P_ε.

Beam search. C3 is anti-monotone under the additive backbone: once all of an item's evidence is masked, further masking cannot restore it. C1 and C2 are not: a retained item can leave the list after one deletion and return after a second. Our beam search (width 5, depth at most 15) searches under P_ε, prunes branches that violate C3, ranks the others by a margin surrogate, and checks the requested constraints at every node. The returned mask is minimal only within the beam.

Heuristics. The removal orders below are heuristics on our backbone, not re-implementations of any published explainer, and we make no claim that a published method reduces to any of them. Each adds at most 15 interactions and stops when the requested setting holds under P_ε. In settings with C3, an addition that would remove the last qualifying evidence of a retained item is skipped.

- *Violation* is dynamic: it adds the interaction that most reduces a hinge surrogate of the constraint violation.
- *Target-score attribution* removes interactions in descending order of A_{it}.
- *Ranking-margin attribution* is dynamic: at each step it removes the interaction with the largest $A_{it} - A_{ir}$, where r is the item currently at rank $K+1$.
- *List-aware* removes in descending order of A_{it} minus the mean contribution of i to the retained items.

- *Popularity* and *random* are controls.

For the target-score order we also run a *C1-stop* mode, which adds interactions until C1 holds and then only reports whether C2 and C3 happen to hold. Two continuous relaxations, a sigmoid and a Gumbel mask, are optimised from an attribution initialisation and then discretised and repaired. They are in-house baselines on this backbone.

Verification. Every mask that any method returns is re-ranked from the model, not from the solver's matrix, and is accepted only when the requested constraints hold under both P and P_ε. A mask that fails is reported as a verification failure and is never counted as a success.

Regression tests. The test suite compares the MILP, with replacement pruning on and off, to exhaustive search of P_ε on synthetic instances with ties and on small instances from the trained model. Those comparisons are not part of the reported samples.

5 Experimental setup

Data. MovieLens-1M [12] ratings of at least 4 are positives. Movies are linked to the Microsoft Satori knowledge graph through the public RippleNet release [28]. Last.fm artist plays are the HetRec 2011 release [5], linked to Satori through the public KGCN release [29]; every play count of at least 1 is a positive. We keep items with a knowledge-graph link, then a core of 10 users and 5 items on MovieLens and of 5 users and 3 items on Last.fm, and we drop entities shared by only one remaining item. Relations with fewer than 200 links on MovieLens, or 20 on Last.fm, are dropped. This leaves 5,646 users and 1,979 movies on MovieLens, all of them still linked, and 1,716 users and 1,286 artists on Last.fm, of which 1,010 keep a link after the relation filter. Each user's positives are shuffled and split 70/10/20 into train, validation and test. Co-occurrence, popularity and the path weights are fit on the training split only. Validation chooses the baseline hyperparameters. Explanations are computed on the training history, which is the input of the deployed model.

Backbone accuracy. Table 1 reports Recall@20 and NDCG@20 on the test split, ranked with the same tie rule as the solver. The backbone is competitive with item-based collaborative filtering and with BPR matrix factorisation on both datasets, so the explanations concern a reasonable recommender rather than a toy one. The learned weight on collaborative paths is $\theta_{\text{co}} = 31.79$ on MovieLens and 42.78 on Last.fm. The popularity weight is $\beta = 0.00$ on MovieLens and 0.48 on Last.fm. The knowledge-graph-only row is a separately trained model with the collaborative relation removed.

Protocol. On each dataset we draw 50 users from each quartile of training-history length (200 users, seed 1) and take every item of each user's top-10 list as a target, giving 2,000 pairs. The main configuration is $K = 10$, $\alpha = 0.5$, $\phi = 0$ and $\varepsilon = 10^{-6}$. Three further MovieLens samples, 25 users per quartile, vary one factor each: the evidence threshold $\alpha \in \{0.25, 0.5, 0.75, 1\}$ under C1 + C2 + C3; the list size $K \in \{5, 10, 20\}$; and the margin $\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}\}$, with exhaustive search wherever $|D_u(t)| \leq 14$. These samples are not the main sample. The exact solver has a budget of 20 seconds. Certification uses 60 seconds per re-solve.

Statistics. Mask-size costs are paired. A difference is computed only on the pairs that are optimal in both compared settings, because the hardest pairs drop out as constraints tighten. Each table

Table 1: Test-set top-20 accuracy (%). MovieLens: 5,646 users. Last.fm: 1,716 users. ItemKNN uses cosine similarity; validation selected the full neighbourhood over neighbourhood sizes all, 50 and 200. BPR-MF has 64 factors and stopped after 10 epochs on the validation split.

	MovieLens-1M		Last.fm	
Model	Recall@20	NDCG@20	Recall@20	NDCG@20
Popularity	20.7	17.6	22.0	10.5
ItemKNN	28.2	25.8	39.5	23.3
BPR-MF	26.7	22.8	31.5	17.8
Backbone, KG paths only	21.8	19.6	22.4	11.8
Backbone, KG and collaborative paths	29.1	26.5	42.2	25.3

Table 2: Certified solver outcomes (%). Each row is 2,000 pairs from 200 users, $K = 10$, $\alpha = 0.5$, $\varepsilon = 10^{-6}$. A mask exists when the outcome is optimal or feasible. The feasible share on MovieLens includes the 28 claimed optima that the second solver did not confirm. The interval resamples users.

Dataset	Setting	Optimal	Feasible	Infeasible	Timeout	Exists [95% CI]
MovieLens-1M	C1	95.30	4.70	0.00	0.00	100.00 [100.00, 100.00]
MovieLens-1M	C1 + C3	83.85	0.75	15.40	0.00	84.60 [82.29, 86.74]
MovieLens-1M	C1 + C2	80.35	2.95	16.55	0.15	83.30 [80.92, 85.75]
MovieLens-1M	C1 + C2 + C3	71.20	1.90	26.90	0.00	73.10 [70.11, 76.05]
Last.fm	C1	95.70	0.00	4.30	0.00	95.70 [94.75, 96.60]
Last.fm	C1 + C3	82.85	0.00	17.15	0.00	82.85 [80.80, 84.75]
Last.fm	C1 + C2	30.05	0.00	69.95	0.00	30.05 [28.05, 31.90]
Last.fm	C1 + C2 + C3	28.05	0.00	71.95	0.00	28.05 [26.15, 30.00]

states how many pairs and users lie behind a difference. All 95% intervals are percentile bootstrap intervals (2,000 replicates) that resample users within each history quartile and keep every target of a resampled user together.

6 Results

6.1 How often is surgical removal possible?

Table 2 is the certified outcome of every main-sample pair. On MovieLens, removing the target alone is always possible: a mask exists for 100.00% of pairs (95.30% with a confirmed minimum, 4.70% with a verified mask whose minimality was not confirmed). Keeping the other nine items lowers this to 83.30% [80.92, 85.75], and keeping their evidence as well lowers it to 73.10% [70.11, 76.05]. The pairs without a mask are proven infeasible (16.55% under C1 + C2, 26.90% under C1 + C2 + C3) except for 0.15% of pairs under C1 + C2, which reached neither a mask nor a proof. On Last.fm, removal itself fails for 4.30% of pairs. Keeping the list is the binding constraint: existence falls to 30.05% [28.05, 31.90], and requiring the evidence as well leaves 28.05% [26.15, 30.00]. Requiring the evidence without the list constraint, C1 + C3, leaves existence at 84.60% on MovieLens and 82.85% on Last.fm, so the evidence constraint is affordable once the target is removed and the list constraint is not. No main-sample pair has a retained item with an empty evidence set. No solve was killed. Median solve times are 2.10, 0.97 and 0.80 seconds for the three MovieLens settings; the longest MovieLens solve took 24.53 seconds.

Table 3: Paired cost of a constraint: difference in certified minimum mask size, on the pairs that are optimal in both settings. The interval resamples users.

Dataset	Comparison	Pairs (users)	Mean [95% CI]	Median	Share > 0
MovieLens-1M	C1+C2 − C1	1535 (198)	1.348 [1.185, 1.524]	0	33.7%
MovieLens-1M	C1+C2+C3 − C1+C2	1401 (198)	0.313 [0.250, 0.377]	0	15.1%
MovieLens-1M	C1+C3 − C1	1598 (200)	0.397 [0.325, 0.479]	0	18.9%
Last.fm	C1+C2 − C1	601 (182)	0.271 [0.212, 0.334]	0	20.0%
Last.fm	C1+C2+C3 − C1+C2	561 (176)	0.007 [0.002, 0.014]	0	0.7%
Last.fm	C1+C3 − C1	1657 (200)	0.020 [0.012, 0.029]	0	1.8%

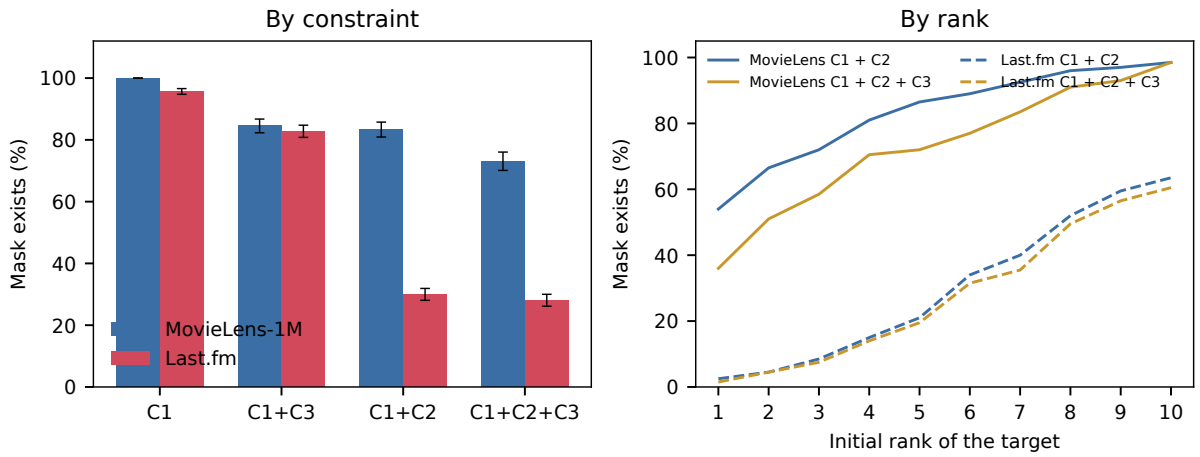


Figure 2: Mask existence after certification. Left: by constraint, with the 95% user-level bootstrap interval. Right: by the target's initial rank, for C1 + C2 (blue) and C1 + C2 + C3 (gold); solid lines are MovieLens-1M and dashed lines are Last.fm. Each point is 200 pairs.

Of the Last.fm pairs with no C1 + C2 + C3 mask, 91.2% are already impossible under C1 + C2. On MovieLens that share is 61.5%; evidence entanglement accounts for 18.2% and a joint conflict, in which C1 + C2 and C1 + C3 are each feasible but not together, for 19.9%. On Last.fm those two causes are 0.4% and 2.4%.

Table 3 is the cost on the pairs where both minima are confirmed. On MovieLens, keeping the list costs 1.348 [1.185, 1.524] extra interactions on average (1535 pairs, 198 users), and keeping the evidence costs another 0.313 [0.250, 0.377] (1401 pairs). Both medians are 0, and only 33.7% and 15.1% of the pairs need any extra interaction, so the averages are carried by a minority of entangled pairs. On Last.fm the same two costs are 0.271 [0.212, 0.334] and 0.007 [0.002, 0.014]. The evidence cost is positive but tiny: once a list-preserving mask exists, one more interaction is rarely required. No pair needs a smaller mask under a stricter setting. Under the minimum masks the retained items reorder only moderately. Mean Kendall's τ between their original and new order is 0.73 and 0.77 on MovieLens and 0.71 and 0.71 on Last.fm.

Rank matters most. Figure 2 and Table 4 show the same rank effect on both datasets. A list-preserving mask exists for 54.0% of MovieLens rank-1 targets and for 98.5% of rank-10 targets (C1 + C2 + C3: 36.0% and 98.5%). On Last.fm the rank-1 rates are 2.5% and 1.5%, against 63.5% and 60.5% at rank 10. A borderline recommendation can usually be removed without side effects.

Table 4: Mask existence (%) by the target's initial rank. Each cell is 200 pairs.

Rank	ML C1+C2	ML C1+C2+C3	LF C1+C2	LF C1+C2+C3
1	54.0	36.0	2.5	1.5
2	66.5	51.0	4.5	4.5
3	72.0	58.5	8.5	7.5
4	81.0	70.5	15.0	14.0
5	86.5	72.0	21.0	19.5
6	89.0	77.0	34.0	31.5
7	92.5	83.5	40.0	35.5
8	96.0	91.0	52.0	49.5
9	97.0	93.0	59.5	56.5
10	98.5	98.5	63.5	60.5

Table 5: Mask existence (%) by training-history quartile, shortest to longest. Each cell is 500 pairs.

History quartile	ML C1+C2	ML C1+C2+C3	LF C1+C2	LF C1+C2+C3
shortest	57.2	42.8	11.8	9.4
second	82.6	66.4	27.2	24.8
third	93.8	86.8	36.0	34.4
longest	99.6	96.4	45.2	43.6

The strongest recommendation usually cannot, and on Last.fm it almost never can.

Short histories are more entangled. Table 5 stratifies by history length. The shortest quartile has the lowest existence under C1 + C2 and under C1 + C2 + C3: 57.2% and 42.8% on MovieLens, against 99.6% and 96.4% in the longest quartile, and 11.8% and 9.4% against 45.2% and 43.6% on Last.fm. With few interactions, each one supports a large part of the list.

6.2 Original problem versus margin variant

Table 6 compares P with P_ε at three margins. The enumerable pairs are 190 pairs from 19 of the 100 sampled users. Feasibility and the optimum never differ, and the MILP agrees with the P_ε enumeration on every decided pair–setting combination. The two problems can differ when a mask needs a gap smaller than ε or an exact tie. On this sample, at these margins, they did not. On the pairs that are too large to enumerate, the indirect check is the one in Table 7: across every stored comparison of an approximate method with the certified optimum, 0 masks were smaller than that optimum and 0 approximate masks succeeded on a pair the MILP had proved infeasible.

6.3 Do constraint-unaware edits preserve the list?

The target-score order that stops at removal reaches C1 for 79.15% of MovieLens pairs and 94.65% of Last.fm pairs. Among those edits, 35.8% and 13.0% keep the other nine items, and 27.5% and 12.4% keep the items and their evidence. Stability is not a by-product of a minimal removal: it has to be asked for. When the same orders are required to satisfy the constraints, beam search and the violation greedy come closest to the exact solver. Beam search returns a verified mask on 63.45% of MovieLens pairs and on 28.05% of Last.fm pairs, which is every Last.fm pair for which a mask

Table 6: P versus P_ε on the MovieLens margin sample, for every pair with $|D_u(t)| \leq 14$. Both problems are solved by exhaustive search. The agreement column sums the three settings, so the denominator is three times the number of pairs. The main samples were not enumerated.

ε	Pairs (users)	Status or optimum differs	MILP agrees
10^{-8}	190 (19)	0	570/570
10^{-6}	190 (19)	0	570/570
10^{-4}	190 (19)	0	570/570

Table 7: Approximate methods under C1 + C2 + C3. Success is the share of all 2,000 pairs for which the verifier accepts the mask. Exact size is the share of those accepted masks, among pairs with a certified optimum, that have that optimum's size. For the stop-at-C1 row, success is the share of pairs whose removal-only mask happens to satisfy C1 + C2 + C3.

	MovieLens-1M		Last.fm	
Method	success	exact size	success	exact size
beam search	63.45	95.90	28.05	100.00
violation greedy	62.05	90.72	27.15	98.89
continuous, sigmoid	71.30	68.52	26.60	97.18
continuous, Gumbel	71.35	63.50	26.30	95.25
ranking-margin order	45.40	75.96	14.35	91.64
list-aware order	44.20	72.18	18.75	93.33
target-score order	32.10	63.56	12.85	95.72
target-score, stop at C1	21.80	76.17	11.75	97.45
popularity order	15.10	37.25	6.45	75.97
random order	15.10	26.49	8.15	77.30

exists, and the mask has the certified size for 95.90% and 100.00% of its successes on proven optima. The violation greedy is close (62.05% and 27.15% success). The repaired continuous masks succeed more often than beam search on MovieLens and less often at the exact size (Table 7). Popularity and random orders rarely satisfy the constraints. No approximate mask in the stored comparisons is smaller than a certified optimum (0), and none succeeds on a pair proved infeasible (0).

6.4 Sensitivity

Table 8 raises the evidence threshold. Mask existence under C1 + C2 + C3 falls from 79.8% at $\alpha = 0.25$ to 69.9% at $\alpha = 1$. The mean size on the pairs that remain optimal does not grow, because the pairs that become infeasible leave the average. Table 9 varies the list. A list-preserving mask exists for 93.20% of pairs at $K = 5$ and 70.95% at $K = 20$ (C1 + C2 + C3: 88.00% and 54.70%). Removal alone is solved for every pair at every K. At $K = 20$, 1.75% of the C1 + C2 pairs time out. Longer lists are harder to preserve because more items share the target's support.

Table 8: Evidence threshold α on a separate MovieLens sample (100 users, 1,000 pairs, C1 + C2 + C3, $K = 10$). Outcomes are percentages. The mean size is over the pairs whose optimum was confirmed, so pairs that become infeasible drop out of it.

α	Exists	Optimal	Infeasible	Mean $\lvert M \rvert$ on optima
0.25	79.8	76.0	20.2	7.19
0.50	76.2	73.0	23.8	6.99
0.75	72.9	70.8	27.1	6.99
1.00	69.9	68.3	30.1	6.92

Table 9: List size K on a separate MovieLens sample (100 users, every top-K item). Outcomes are percentages. The $K = 10$ row is not the main sample.

	C1		C1 + C2		C1 + C2 + C3	
K	exists	timeout	exists	timeout	exists	timeout
5	100.00	0.00	93.20	0.00	88.00	0.00
10	100.00	0.00	84.70	0.20	74.90	0.10
20	100.00	0.00	70.95	1.75	54.70	0.00

7 Discussion and limitations

What the numbers mean. On MovieLens, most recommendations, and almost all borderline ones, have a small edit that removes the item and leaves the rest of the list and its evidence intact. The strongest recommendations, and the users with the shortest histories, often do not. On Last.fm the list constraint binds much earlier: a list-preserving edit exists for about a third of pairs, and for the top-ranked item almost never. In both datasets the failures are mostly proved, not merely unsolved. The exact formulation turns that entanglement into a measured quantity, and it gives explainers a ground truth. The ground truth is the certified optimum: one MovieLens optimum of size 4 was replaced by a verified optimum of size 3, and 28 claimed MovieLens optima remain feasible rather than optimal because the second solver did not confirm them.

Limitations. (i) The backbone is additive by design, so that path contributions are exact and C3 can be checked in closed form. Non-additive recommenders need a single-path-disabling definition of contribution and an approximate solver; this is study 2. (ii) The exact solver works on P_ε. We compared it with P only on the 190 enumerable pairs of the margin sample, which come from 19 users and mostly from short histories. (iii) The sensitivity samples have 100 users and are separate from the main 200. (iv) The heuristics, the beam and the continuous masks are our own procedures on this backbone. PRINCE was not run at this scale, and the official implementations of CoSRec, GREASE, LXR, SPINRec, ACCENT and TRACE have not been run here, so no statement is made about how those methods would perform under C2 and C3. (v) The knowledge graphs are the Satori linkages released with RippleNet and KGCN, not KB4Rec or Freebase. (vi) The literature check was focused rather than systematic, so the joint guarantee is framed as the proposed contribution and the priority claim is left open. (vii) The backbone is static. User preferences drift over time [23, 24]; with time-dependent scores the score row of an interaction would change between requests, and extending evidence-preserving counterfactuals to such dynamic recommenders is future work.

8 Conclusion

EPIC-Rec asks for counterfactual explanations that remove one recommendation without collateral damage to the rest of the list or to its evidence. On an additive path recommender the question has an exact, certified answer. On MovieLens that answer is usually yes, except for the items the user is most strongly recommended. On Last.fm, keeping the rest of the list is possible for only a minority of recommendations. The formulation and the exact solver give a test bed for explainers that claim list-level faithfulness.

Code and data availability. The implementation and the certified result files are in the EPIC-Rec repository. Every number and table in this paper is written from those files by `scripts/manuscript_from_report.py`. Raw datasets are not redistributed.

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